

WEEK 10: PROBABILITY

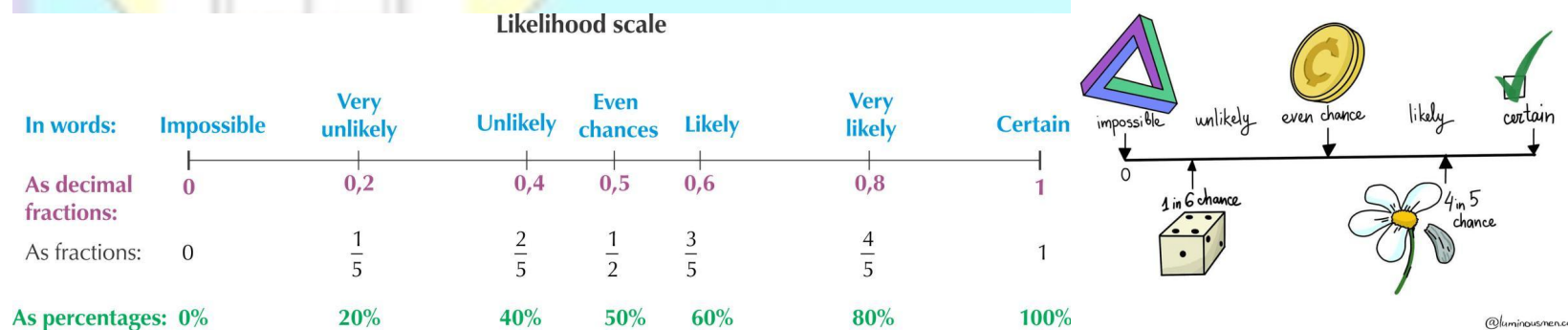
A. Meaning of Probability

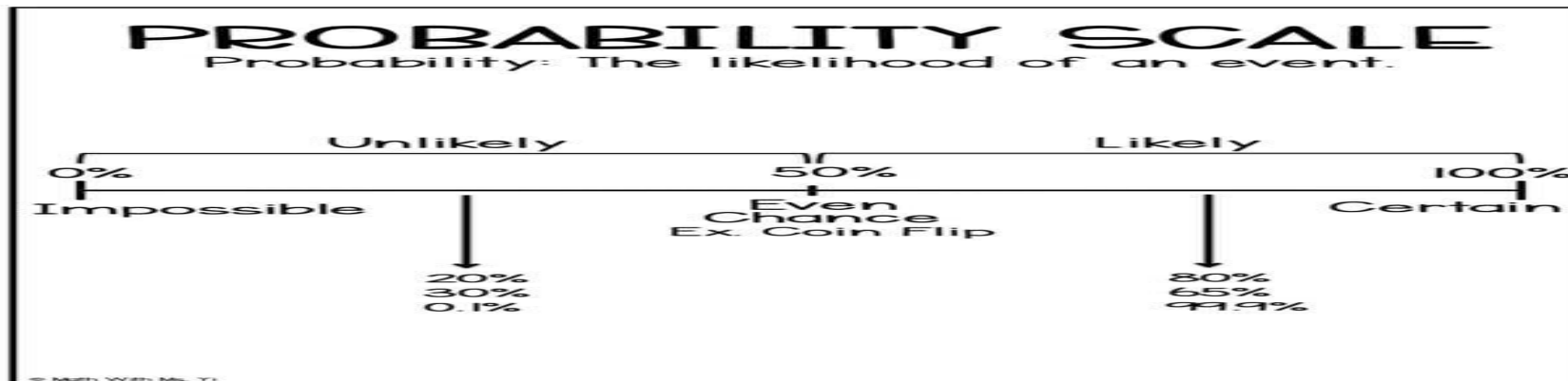
Definition (Short Note)

Probability measures how likely an event is to happen. I.e the possibility

- Values range from **0 to 1**
- Can also be written as **0% to 100%**

Probability Scale





5

Interpretation

- 0 → Impossible
- 0.5 → Equal chance
- 1 → Certain

Basic Formula

$$P(E) = n(E) / n(S)$$

Where:

- $P(E)$ = probability of event
- $n(E)$ = favorable outcomes
- $n(S)$ = total outcomes



Key Properties

- $0 \leq P(E) \leq 1$
- Total probability = 1
- Impossible event = 0
- Certain event = 1

B. Sample Space

Definition (Short Note)

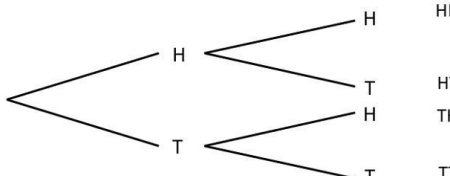
Sample space is the **set of all possible outcomes**.

Examples

- Coin: {H, T}
- Die: {1,2,3,4,5,6}
- Two coins: {HH, HT, TH, TT}

Visualizing Sample Space

1st Coin 2nd Coin



HH
HT
TH
TT

Outcomes for Two Dice

	1	2	3	4	5	6
1	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
2	(2, 1)	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
3	(3, 1)	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
4	(4, 1)	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
5	(5, 1)	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
6	(6, 1)	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

Player 2

	1	2	3	4	5	6
1	1,1	1,2	1,3	1,4	1,5	1,6
2	2,1	2,2	2,3	2,4	2,5	2,6
3	3,1	3,2	3,3	3,4	3,5	3,6
4	4,1	4,2	4,3	4,4	4,5	4,6
5	5,1	5,2	5,3	5,4	5,5	5,6
6	6,1	6,2	6,3	6,4	6,5	6,6

Player 1

Ways to List Sample Space

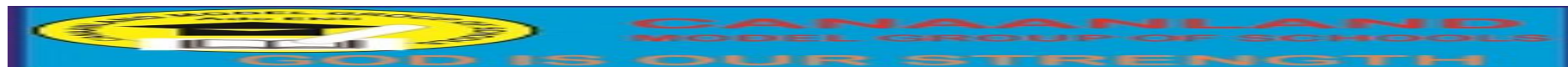
1. Tree Diagram
2. Table/Grid Method

C. Types of Events

1. Simple Event

One outcome only

EXAMPLES:



- Rolling a 4 on a die: {4}

- Drawing the Ace of Spades: {A♠}
- Getting exactly 2 heads in 3 tosses: {HHT, HTH, THH}. Probability of (Rolling a 4) = $\frac{1}{6}$

2. Compound Event

More than one outcome i.e An event that consists of two or more simple events

EXAMPLES: • Rolling an even number: {2, 4, 6} • Drawing a red card: {All hearts and diamonds} • Getting at least 1 head in 2 tosses: {HH, HT, TH}

3. Mutually Exclusive Events

Meaning (Short Note)

Events that cannot happen at the same time

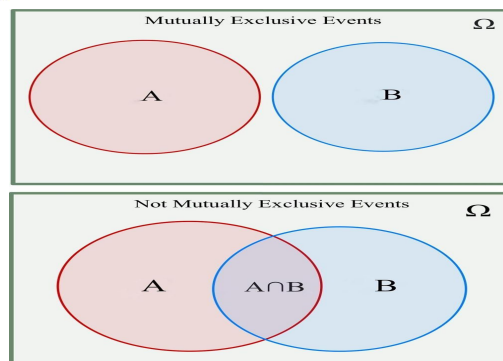
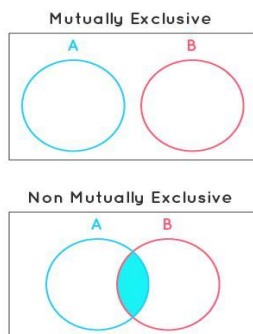
EXAMPLES: • Rolling a 2 AND rolling a 5 (one die)

- Turning left AND turning right

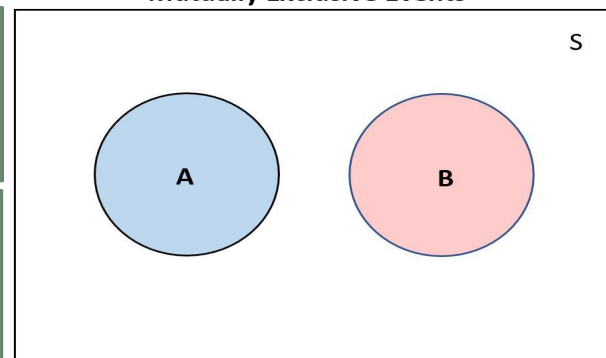
- Being absent AND being present

Visual Illustration

Venn Diagram Showing Mutually Exclusive Event



Mutually Exclusive Events



Formula

$$P(A \text{ or } B) = P(A) + P(B)$$

4. Non-Mutually Exclusive Events

Events can happen together

Example of Non-Mutually Exclusive Events:

• Drawing a King OR a Heart (King of Hearts is both!) • Being tall OR being smart • Rolling even OR rolling > 3 {2,4,6} and {4,5,6} overlap at 4,6

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

5. Independent Events

Meaning



One event **does not** affect the other

Examples of independent Events • Tossing coin twice (First toss doesn't affect second) • Rolling two dice • Drawing with replacement

Formula:

For independent events: $P(A \text{ and } B) = P(A) \times P(B)$

EXAMPLE: $P(\text{Heads then Tails}) = P(H) \times P(T) = 1/2 \times 1/2 = 1/4$

6. Dependent Events

Events where the outcome of one DOES affect the outcome of the other

EXAMPLES: • Drawing cards without replacement • Selecting people from a group • Taking items from a bag without returning

$P(A \text{ and } B) = P(A) \times P(B|A)$ Where $P(B|A)$ = Probability of B given A happened

7. Complementary Events

Meaning

Opposite events

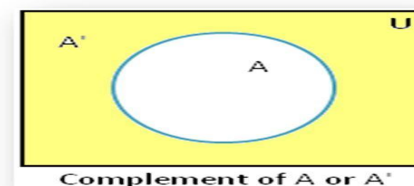
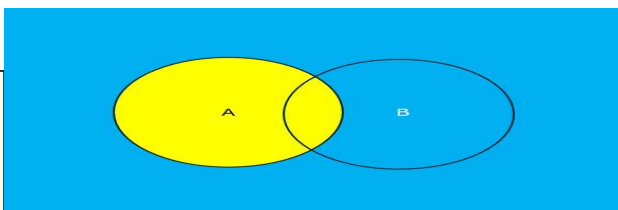
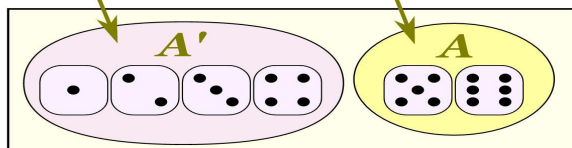
Example:

- Pass / Fail
- Rain / No rain

Illustration

Complement of Event A

Event A



Complement of A or A'

Formula

$$P(A) + P(A') = 1 \quad P(A') = 1 - P(A)$$

Worked Examples (Simplified)

Example 1 (Balls): A bag contains 5 red balls, 3 blue balls, and 2 green balls. A ball is drawn at random. Find the probability that it is:
 (a) Red

Total = 10

- $P(\text{Red}) = 5/10 = 1/2$
- $P(\text{Blue}) = 3/10$
- $P(\text{Not green}) = 4/5$

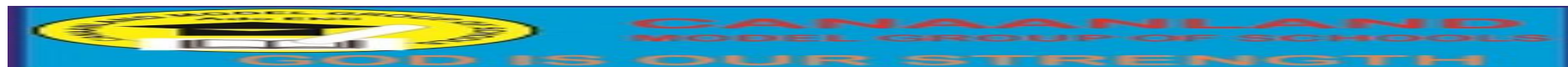
Example 2 (Die) A die is rolled once. Find the probability of getting: (a) An even number (b) A number greater than 4 (c) An even number or a number greater than 4 :

Solution: Sample space = {1, 2, 3, 4, 5, 6}, $n(S) = 6$ (a)

Even numbers = {2, 4, 6} $P(\text{Even}) = 3/6 = 1/2$ (b)

Numbers > 4 = {5, 6} $P(>4) = 2/6 = 1/3$

(c) Even OR >4 = {2, 4, 5, 6} These are NOT mutually exclusive (6 is both)



$$P(\text{Even or } >4) = P(\text{Even}) + P(>4) - P(\text{Even and } >4) = 3/6 + 2/6 - 1/6 = 4/6 = 2/3$$

Example 3 (Coins) Two coins are tossed. Find the probability of getting: (a) Two heads (b) At least one head (c) No heads

Sample space = {HH, HT, TH, TT}

- $P(2 \text{ heads}) = 1/4$
- $P(\text{at least 1 head}) = 3/4$
- (c) No heads = {TT} $P(\text{No heads}) = 1/4$

Example 4 (Cards) A card is drawn from a standard deck of 52 cards. Find the probability that it is: (a) A King (b) A Heart (c) A red card (d) A King or a Heart

Solution:

(a) $P(\text{King}) = 4/52 = 1/13$

(b) $P(\text{Heart}) = 13/52 = 1/4$

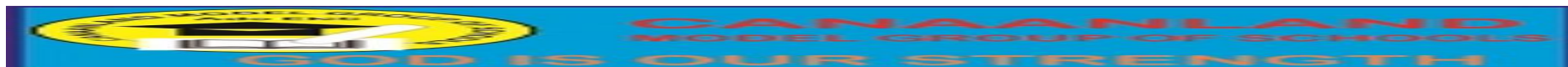
(c) $P(\text{Red}) = 26/52 = 1/2$

(d) $P(\text{King or Heart})$: These are NOT mutually exclusive (King of Hearts exists) $P(\text{King or Heart}) = P(\text{King}) + P(\text{Heart}) - P(\text{King and Heart}) = 4/52 + 13/52 - 1/52 = 16/52 = 4/13$

(e) Example 5 (Without Replacement): A bag contains 6 red marbles and 4 blue marbles. Two marbles are drawn one after another without replacement. Find the probability that: (a) Both are red (b) Both are blue (c) One is red and one is blue

Solution: Total marbles = 10

- Both red = $1/3$
- Both blue = $2/15$



- One red, one blue = $8/15$

Example 6 (Complement): The probability that it will rain tomorrow is 0.3. What is the probability that it will not rain?

$P(\text{Not rain}) = 0.7$

Example 7: A fair coin is tossed three times. Find the probability of getting exactly two heads.

Solution: Sample space for 3 tosses: {HHH, HHT, HTH, HTT, THH, THT, TTH, TTT} $n(S) = 8$ Exactly 2 heads = {HHT, HTH, THH}
 $n(\text{Exactly 2H}) = 3$ $P(\text{Exactly 2H}) = 3/8$

Example 8: In a class of 30 students, 18 study Mathematics, 15 study Physics, and 10 study both. If a student is selected at random, find the probability that the student studies: (a) Mathematics or Physics (b) Neither Mathematics nor Physics

Solution: (a) $P(M \text{ or } P) = P(M) + P(P) - P(M \text{ and } P) = 18/30 + 15/30 - 10/30 = 23/30$ (b) $P(\text{Neither}) = 1 - P(M \text{ or } P) = 1 - 23/30 = 7/30$

Example 9: Two dice are rolled. Find the probability that: (a) The sum is 7 (b) The sum is greater than 9 (c) Both dice show the same number

Solution: Total outcomes = $6 \times 6 = 36$ (a) Sum = 7: {(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)} $n = 6$ $P(\text{Sum} = 7) = 6/36 = 1/6$

(b) Sum > 9 means sum = 10, 11, or 12 Sum = 10: {(4,6), (5,5), (6,4)} = 3 Sum = 11: {(5,6), (6,5)} = 2 Sum = 12: {(6,6)} = 1 Total = 6
 $P(\text{Sum} > 9) = 6/36 = 1/6$

(c) Same number: {(1,1), (2,2), (3,3), (4,4), (5,5), (6,6)}

Example 10: A spinner has sectors numbered 1 to 8. If the probability of landing on an even number is 0.5, how many even-numbered sectors are there? **Solution:** $P(\text{Even}) = \text{Number of even sectors} / \text{Total sectors}$ $0.5 = n/8$ $n = 0.5 \times 8 = 4$ There are 4 even-numbered sectors

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- **Classwork:** 1. Define probability and state its range of values.
 - What is a sample space? Give an example.



- . A bag contains 5 black and 3 white balls. Find $P(\text{black})$.
- . Two events A and B are mutually exclusive. If $P(A) = 0.3$ and $P(B) = 0.4$, find $P(A \text{ or } B)$.
- . If $P(E) = 0.65$, find $P(E')$.

Assignment:

1. . A die is rolled once. Find the probability of: (a) Rolling a 5 (b) Rolling an odd number (c) Rolling a number less than 4 (d) Not rolling a 6
2. A box contains 4 red, 5 blue, and 6 green marbles. A marble is drawn at random. Find: (a) $P(\text{Red})$ (b) $P(\text{Not blue})$ (c) $P(\text{Red or green})$
3. Two coins are tossed. List the sample space and find the probability of: (a) Getting two tails (b) Getting at least one head (c) Getting exactly one tail
4. A card is drawn from a standard deck. Find: (a) $P(\text{Ace})$ (b) $P(\text{Club})$ (c) $P(\text{Red King})$ (d) $P(\text{Ace or Club})$
5. Explain the difference between: (a) Mutually exclusive and independent events (b) Complementary events and mutually exclusive events
Give an example for each.

